

# Algorithms and Applications

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## Engineering Technology

May, 2026



# Algorithms and Applications (A13122)

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|---------------------|------------------------------|------------------------|
| <b>Short Title:</b> | Algorithms & Applications    |                        |
| <b>Faculty</b>      | Science and Computing        |                        |
| <b>Department</b>   | Engineering Technology       |                        |
| <b>Cluster</b>      | Software and Web Development |                        |
| <b>Author</b>       | CLAIRE.FITZPATRICK           |                        |
| <b>Credits</b>      | 5                            | <b>Level:</b> Advanced |

## Description of Module / Aims

The aim of this module is to develop the ability of the student to understand, apply the mathematics required, develop the algorithms, program and evaluate topics in machine learning, artificial intelligence, and power systems. Topics are selected based on current research and publications and are selected to be as engaging as possible for the student.

## Programmes

|           |                                                                                       |           |
|-----------|---------------------------------------------------------------------------------------|-----------|
| MTHE-0061 | BEng (Hons) in Electrical Engineering (WD_ETRIC_B)                                    | 4 / 7 / M |
|           | BEng (Hons) in Electrical and Automation Engineering (International)<br>(WD_ETRIC_BI) | 4 / 7 / M |
| MTHE-0061 | BEng (Hons) (WD_ETRIC_B)                                                              | 4 / 7 / M |

## Indicative Content

- Review of programming methods; A selection of topics using any of the following typical methods to determine Machine Learning, Artificial Intelligence and power systems problems
- The development of algorithms for particular problems in Machine Learning, Artificial Intelligence and power systems are changing rapidly. It is envisaged that the content will be dynamic based on the interest of the students.

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Analyse problems to develop structured algorithms for various Machine Learning, Artificial Intelligence and power systems intelligent applications.
2. Perform the required mathematics for various Machine Learning, Artificial Intelligence and power systems intelligent applications.
3. Program and integrate a combination of data flow, functional and numerical computing environment languages with some typical devices used in electrical engineering to maximize the development speed and performance for various Machine Learning.

## Learning and Teaching Methods

- Lecturer led combined practicals / lectures
- Students continuous engagement with the course material is encouraged by providing the course materials, examples and software to the student in advance.
- Student's timely engagement with the course material is encouraged by providing clear criteria for proportionate and timely assignments and assessments.
- Peer to peer learning using group assignments and assessments
- Timely self-directed learning, in addition to lectures/practicals, that is required by the student to develop programming ability

## Assessment Methods

|                       | Weighing | Outcomes Assessed |
|-----------------------|----------|-------------------|
| Continuous Assessment | 100%     |                   |
| Group Project         | 15%      | 1,2               |
| Group Project         | 15%      | 3                 |
| Assignment            | 30%      | 1,2               |
| Assignment            | 40%      | 3                 |

## Assessment Criteria

**70% -**

**100%:** The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

**60% - 69%:** The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

**50% - 59%:** The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

**40% - 49%:** The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

**<40%:** The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Practical            | 48        |           |
| Independent Learning | 87        |           |

## Essential Material

- National Instruments Online. \_Getting Started with NI LabView Student Training\_ .. 2013.

## Requested Resources

- Room Type: Computer Lab